The Effect of Epigallocatechin-3-Gallate (EGCG) on Cell viabity of the Gastric Cancer Cell Line KATO-III

Introduction

GASTRIC CANCER

AND THE USE OF EPIGALLOCATECHIN-3-GALLATE

Malignant tumor that begins in the inner lining (mucosa) of the stomach, developing when stomach cells grow uncontrollably and form tumors. Most gastric cancers are adenocarcinomas, originating from mucus-producing cells (National Cancer Institute, n.d.).

1 MILLION estimated cases annually

5th Gastric cancer is the most 5th most common Cancer

3rd The third leading cause of cancer deaths globally as of 2020

TREATMENT CHALLANGES

- Tumors over time begin to resist chemotherapy, radiation, and hormonal therapy by avoiding aptosis
- Gastric Cancer must be caught early and the use of alternative therapies

SIGNS

Weight loss

Nausea

Abdominal Pain

(Smyth et al., 2020)

Risk Factors (Smyth et al., 2020)

Old Age

Diets low in fruits and vegitables

High Salt Intake

Helicobacter pylori infection

EPIGALLOCATECHIN-3-GALLATE

- Potent antioxidant extracted from green tea
- In a study conducted by Zhu et al. (2007), they studied the effects of EGCG on the volume, weight, and size of a gastric cancer tumor in which they observed a significant reduction across all areas

Project Overview

Research Question

Can EGCG reduce the cell viability of gastric cancer?

Hypothesis

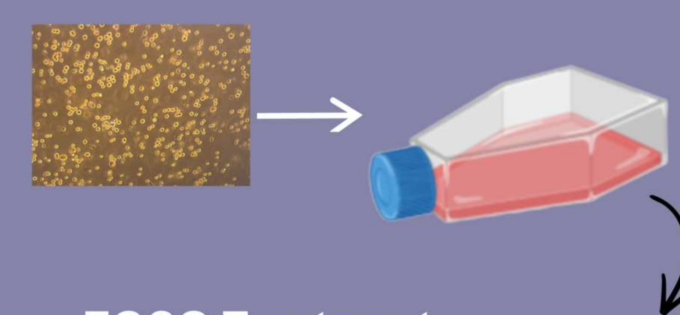
If KATO-III cells are treated with concentrations of EGCG, then as the concentration increases there will be reduction in the cell viability.

Justification

- EGCG is known to **initiate apoptosis** in colon cancer at a concentration of 100μM.
- In a study conducted by Zhu, they studied the effects of EGCG on the volume, weight, and size of a gastric cancer tumor in which they observed a **significant reduction** in across all areas.
- With this knowledge, when working with KATO-III cells the **cell viability should decrease** supporting Zhu's study.

Methodology

Culture KATO-III



EGCG Treatment

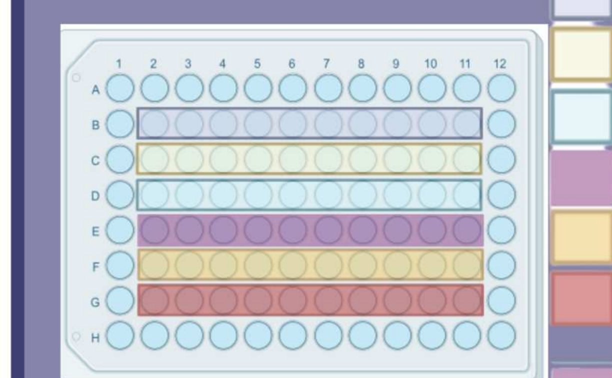


Figure 1: 96-well plate depicting the experimental setup of EGCG treatment and the content of each well

1mM EGCG

0.5mM EGCG

0.25mM EGCG

0mM EGCG + DMSO (Vehicle Control)

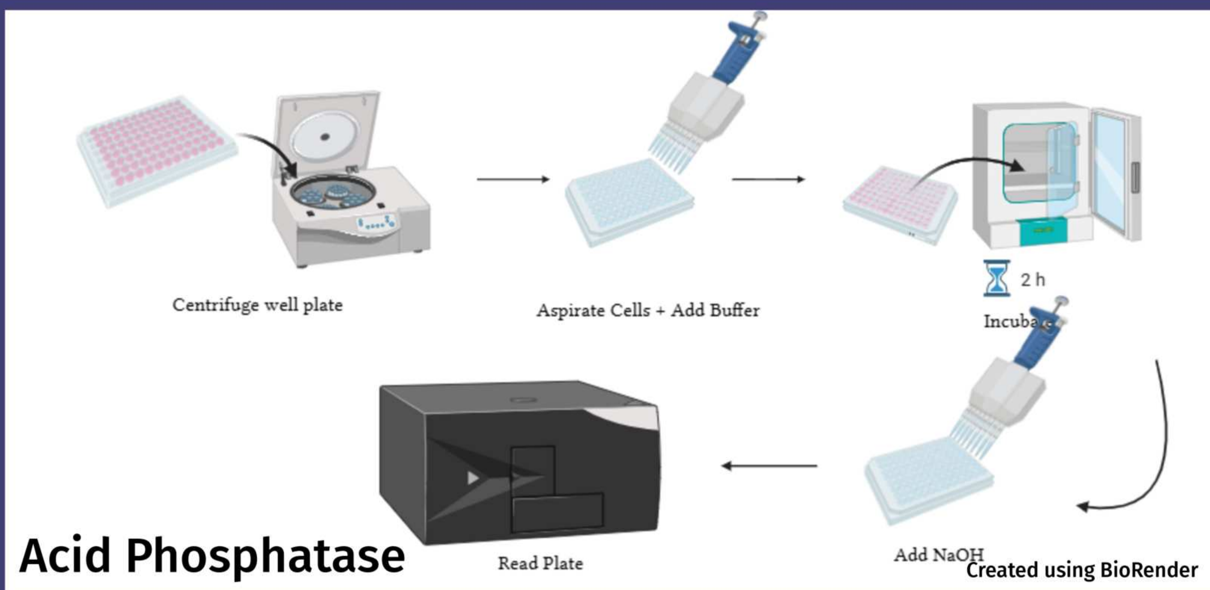
and

Complete Media + Cells (Negative Control)

*Seperate plates

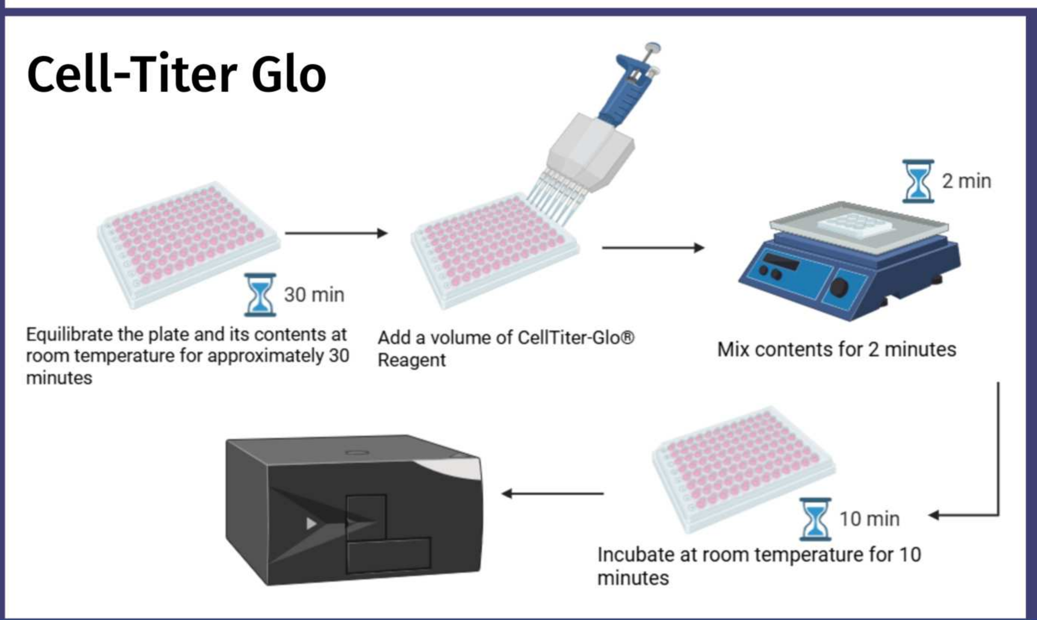
Incubation

48 HOURS × 37 °C



Acid Phosphatase

Cell-Titer Glo



Results

Cell Titer Glo

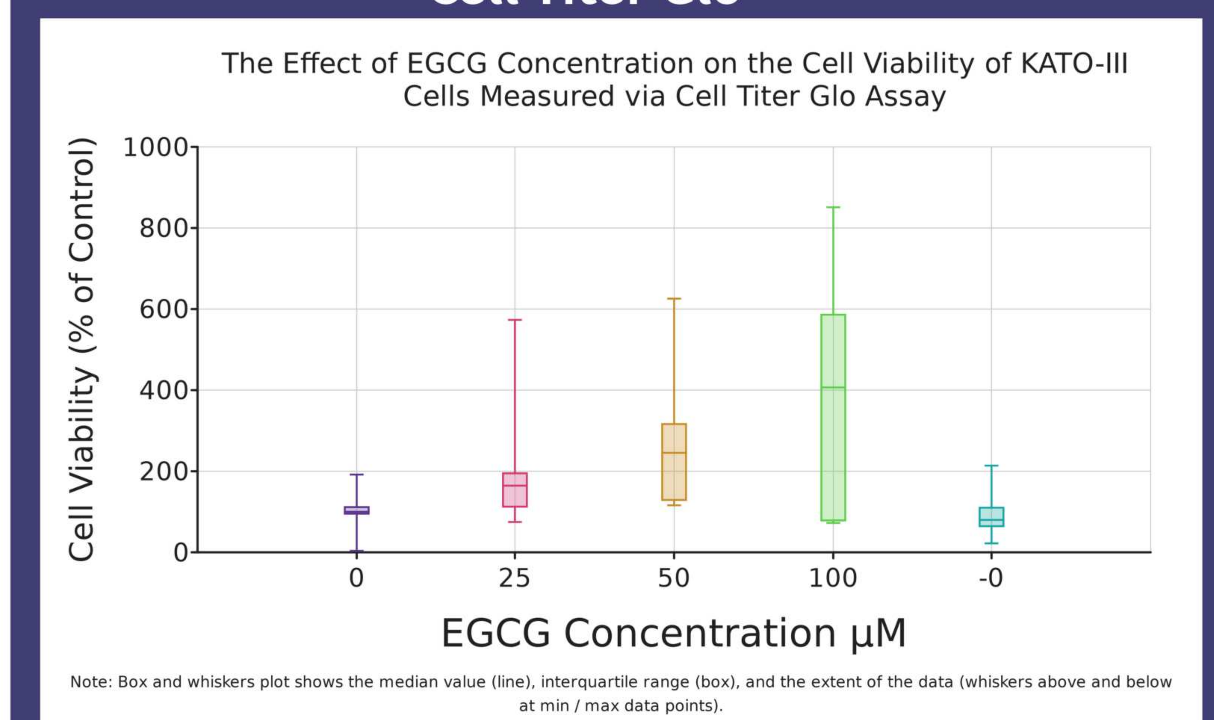


Figure 2: Represents the the relationship between the EGCG concentration and the cell viability of the gastric cancer cell line KATO_III. 0 represents the negative contorl group of cells+media while -0 represnts the vehicle control

Kruskal-

Degrees of Freedom (df)	P-value	Interpretation of P
4	31.58	A P-value of **<0.01** means **extremely strong confidence** that the groups are different.

Table 1: Represents results of the cinducted Kruskal-Wallis statistical test.

Post-Hoc Test

Table 2: The table represents the results of the post-hoc test. Stars represent the reults in whihc the researcher foucsed on and the results important to drawing a conclusion.

Pair	P-value		Interpretation of P
0 - 25	1.00	★	Means: that the groups are **NOT** different
0 - 50	1.00	★	Means: that the groups are **NOT** different
0 - 100	<0.01	★	Means: **extremely strong confidence** that the groups are different
0 - -0	0.03		Means: **strong confidence** that the groups are different
25 - 50	1.00		Means: that the groups are **NOT** different
25 - 100	<0.01		Means: **extremely strong confidence** that the groups are different
25 - -0	<0.01	★	Means: **extremely strong confidence** that the groups are different
50 - 100	0.01		Means: **very strong confidence** that the groups are different
50 - -0	0.11	★	Means: **some indication** that the groups **might** be different
100 - -0	1.00	★	Means: that the groups are **NOT** different

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Acid Phosphatase

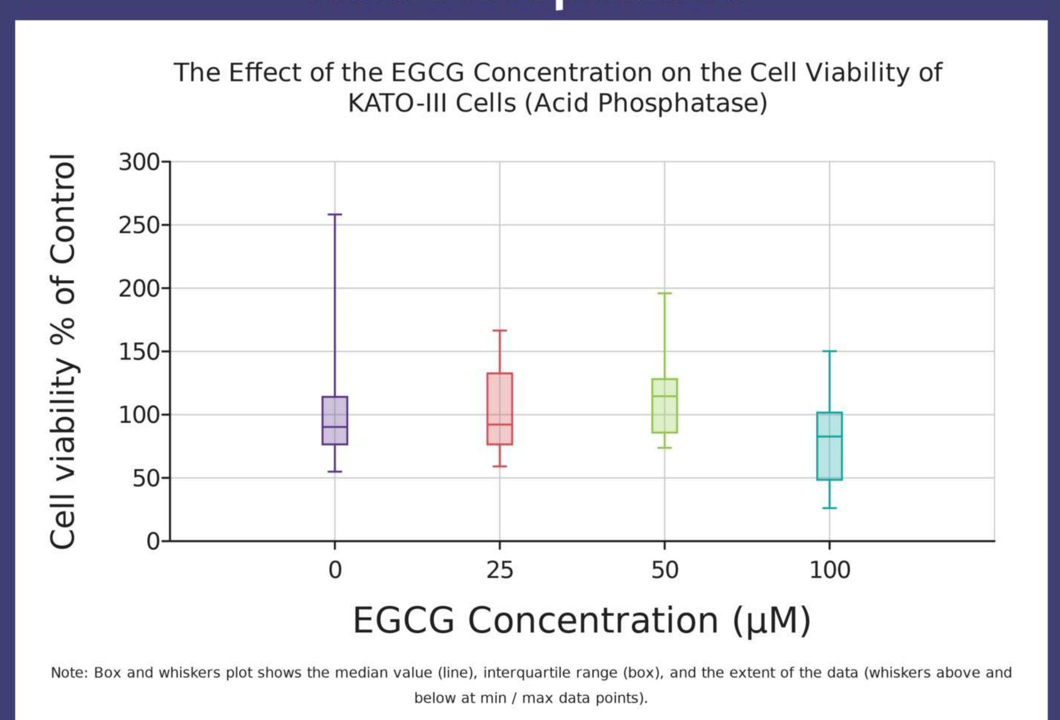

Figure 5: Represents the relationship between the EGCG concentration and the cell viability of the gastric cancer cell line KATO_III using the results of the acid phosphatase assay. 0 represnts the vehicle control

Kruskal-

Degrees of Freedom (df)	P-value	Interpretation of P
3	0.07	A P-value of **0.07** means **some confidence** that the groups are different.

Table 3: Represents results of the cinducted Kruskal-Wallis statistical test.

Post-Hoc Test

Pair		P-value	Interpretation of P
0 - 25	★	0.06	Means: **some confidence** that the groups are different
0 - 50	★	0.66	Means: **no evidence** that the groups might be different
0 - 100	★	0.91	Means: **no evidence** that the groups might be different
25 - 50		1.00	Means: that the groups are **NOT** different
25 - 100		0.50	Means: **little to no indication** that the groups might be different
50 - 100		1.00	Means: that the groups are **NOT** different

Table 4: The table represents the results of the post-hoc test. Stars represent the reults in whihc the researcher foucsed on and the reulsts important to drawing a conclusion.

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Discussion

Cell Titer Glo

- **Results do not support hypothesis nor past reasech regarding the effect of EGCG on gastric cancer cells**
- **Cell viability increased** with the rising EGCG concentration
- The Kruskal–Wallis test indicated a **statistically significant difference between groups** ($p < 0.01$), suggesting that at least one concentration differs from the others. Which was supported with the post-hoc test which determined the **100 µM EGCG concentration had a significant difference to the negative control (0).**
- Results don't resemble expected results apoptosis at an EGCG concentration of 100 µM
- EGCG is known to interact with cellular metabolism and redox reactions, which may artificially affect ATP-based luminescence readings in this assay.

Acid Phosphatase

- Results **support some of the hypothesis**
- Cell viability was reduced at an EGCG concentration of 100 µM suggesting a potential cytotoxic effect at higher EGCG concentrations.
- Cell viability remained relatively stable across 25, and 50 µM, with only slight increases above control levels.
- The Kruskal–Wallis test produced a p-value of 0.07, indicating that differences between groups were not statistically significant, but show a trend toward significance.
- Post-hoc showed **no significant differences**, although the comparison between 0 µM and 25 µM ($p = 0.06$) suggests a **possible mild effect at low concentrations**.
- Results suggest **low concentrations of EGCG slightly enhance cell activity while higher concentrations inhibit it.**
- Acid phosphatase activity is less likely to be affected by metabolic interference → **results more biologically reliable**.
- Overall, the acid phosphatase assay provides weaker statistical evidence but more realistic biological trends, suggesting that EGCG has minimal effect at low doses and potential toxicity at high doses.

Future Work

The current findings provide an understand that EGCG is unable to reduce the cell viability of KATO-III cells. Using these results the RNA-binding protein, SERBP1, can be observed in future work to determing how EGCG effects the protein SERBP1.

- **SERBP1 is a RNA-binding protein aids in the growth and migration of gastric cancer cells**
- **Possible corrections that can be made in future work is including both controls on one well plte rather than two seperate 96-well plates**

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